

SET

A

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE: CSE4125 (Distributed Database Systems)
 Spring 2018 | Quiz – 3 | Marks 20 | Time: 30 Minutes

ID: _____

Section: _____

Consider the following global and fragmentation schemata, and a global query.

Global schema: **STUDENT** (SNUM, SNAME, DNAME, SEM)
DEPT (DNAME, HEAD, PROGRAM)

Fragmentation schema: **STUDENT**₁ = SL SEM ≥ 4 and DNAME = "EEE" STUDENT
STUDENT₂ = SL SEM ≥ 4 and DNAME = "CSE" STUDENT

DEPT₁ = SL DNAME = "EEE" PJ DNAME, PROGRAM DEPT

DEPT₂ = SL DNAME = "CSE" PJ DNAME, PROGRAM DEPT

DEPT₃ = PJ DNAME, HEAD DEPT

(Assume that "CSE" and "EEE" are the only possible values for DNAME)

Global query: (SL SEM < 4 **STUDENT**) UN (SL DNAME = "CSE" **DEPT** NJN SL PROGRAM = "MSC" **DEPT**) UN (SL DNAME = "CSE" **STUDENT** DF SL DNAME ≠ "EEE" **STUDENT**)

Determine the equivalent *simplified* fragment query for the above global query. Indicate rules and criteria applied in your steps.

SET

B

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Consider the following global and fragmentation schemata, and a global query.

Global schema: **STUDENT** (SNUM, SNAME, DNAME, SEM)
DEPT (DNAME, HEAD, PROGRAM)

Fragmentation schema: **STUDENT**₁ = SL SEM < 4 and DNAME = "CSE" STUDENT
STUDENT₂ = SL SEM < 4 and DNAME = "ME" STUDENT

DEPT₁ = SL DNAME = "CSE" PJ DNAME, PROGRAM DEPT

DEPT₂ = SL DNAME = "ME" PJ DNAME, PROGRAM DEPT

DEPT₃ = PJ DNAME, HEAD DEPT

(Assume that "CSE" and "ME" are the only possible values for DNAME)

Global query: (SL SEM ≥ 4 **STUDENT**) UN (SL DNAME = "CSE" **DEPT** NJN SL PROGRAM = "BSC" **DEPT**) UN (SL DNAME = "CSE" **STUDENT** DF SL DNAME ≠ "ME" **STUDENT**)

Determine the equivalent *simplified* fragment query for the above global query. Indicate rules and criteria applied in your steps.